Polynomial-Resource Classification of Quantum Circuit Families via Classical Shadows

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***Abstract*—We compare four polynomial-resource measurement strategies, (I) Z-basis-only, (II) nearest-neighbor ZZ (NN), (III) multi-basis (Z, X, Y), and (IV) classical shadows, for classifying three quantum circuit families: IQP, Clifford, and Clifford+T. We find Z-only measurements outperform multi-basis and classical shadows across all qubit counts and all four classifiers evaluated, and the $O(n)$-feature NN strategy matches Z-only to within 0.02 in Random Forest accuracy. The best result is a Random Forest accuracy of 0.91 at 4–5 qubits under Z-only (0.89 for NN, 0.85 for multi-basis, 0.67 for shadows). All four strategies collapse to near-chance accuracy (≈ 0.33) above approximately 12 qubits under the quadratic shot budget $s = 16n^2$. These findings indicate that the discriminative signal between these circuit families is concentrated in local, nearest-neighbor Z-basis correlations, consistent with the diagonal gate structure of IQP circuits, and that additional Pauli correlator types or long-range correlations carry no compensating discriminative power for this task. We provide a formal theoretical framework showing that circuits with high diagonal fraction in a given basis concentrate their correlator structure in that basis, and that any deviation from the dominant basis incurs a provably higher estimator variance. These results establish that a quadratic shot budget is insufficient for reliable classification above approximately 12 qubits, but do not rule out the existence of a subquadratic or otherwise more efficient polynomial-resource strategy; whether any polynomial measurement protocol can classify these families at large qubit counts remains an open question.**

***Index Terms*—classical shadows, quantum circuit families, IQP circuits, Clifford circuits, polynomial resources, distribution classification, Pauli correlators, quantum machine learning**

I. Introduction

A fundamental question in quantum information is whether the output distributions of quantum circuits, distributions that cannot be efficiently represented classically, can be meaningfully characterized using only polynomially many measurements [1]. Quantum state tomography, originally proposed by Vogel and Risken (1989) [2] and first demonstrated experimentally by Smithey et al. (1993) [3], provides a complete reconstruction of the quantum state but at prohibitive cost. Haah et al. (2016) [4] rigorously established this cost with a lower bound of $\Omega(d^2/\delta)$ copies, where d is the Hilbert space dimension and δ is the desired precision. Since $d = 2^n$ for an n-qubit system, full tomography requires exponentially many measurements in the number of qubits, rendering it infeasible for systems of practical interest. This exponential barrier motivates the search for protocols that extract meaningful information about quantum states without full reconstruction.

Quantum circuit families provide a principled testbed for this question. IQP, Clifford, and Clifford+T circuits produce distributions with qualitatively different statistical signatures, rooted in their distinct computational complexity properties. Clifford circuits are classically simulable; IQP sampling is believed to be classically hard under plausible complexity-theoretic conjectures [5]; Clifford+T circuits, a universal gate set [6], [7], interpolate between the two. If these families can be distinguished with polynomially many measurements, it suggests that their distributional signatures are encoded in low-order statistics that a polynomial measurement protocol can access.

The key methodological challenge is choosing the right measurement protocol. Measuring in a single basis (Z-only) gives access to ZZ correlators but misses the X and Y components of the Pauli correlation structure. Measuring in three global bases (Z, X, Y) recovers ZZ, XX, and YY correlators but requires $3\times$ the shot budget and still misses mixed-basis correlators (XY, XZ, YZ). Classical shadows resolve this. By applying a random single-qubit Clifford rotation before measurement, all six two-qubit Pauli correlators can be estimated from a single unified protocol with sample complexity $O(\log n^2) = O(\log n)$ to predict all pairwise correlators simultaneously, significantly more efficient than the $O(n^2)$ shots required to estimate each correlator type independently. In this work, all four measurement strategies

share the same $O(n^2)$ shot budget to ensure a fair experimental comparison; under this budget classical shadows accesses all six Pauli correlator types simultaneously, while Z-only, NN, and multi-basis access fewer.

Contributions:

- An empirical comparison of four measurement strategies (Z-only, nearest-neighbor ZZ, multi-basis, and classical shadows) for classifying IQP, Clifford, and Clifford+T circuit families, showing that Z-only and NN outperform multi-basis and shadows and inverting the hypothesis that richer Pauli correlation structure would benefit classification.
- Evidence that the discriminative signal is concentrated in local, nearest-neighbor ZZ correlations: an $O(n)$-feature NN strategy matches the $O(n^2)$-feature Z-only strategy to within 0.02 in Random Forest accuracy, consistent with the diagonal computational-basis structure of IQP circuits.
- Characterization of a scaling collapse near 12 qubits at which all implemented polynomial strategies degrade to near-chance performance under the quadratic shot budget, establishing a practical limit for this feature set and shot scaling law.
- A formal theoretical framework (Theorem 3, Proposition 5, and Corollary 8) showing that the empirical superiority of Z-only measurement is a provable consequence of the diagonal gate structure of IQP circuits, not merely an empirical artifact.

II. Background

A. Circuit Families as Distribution Classes

We consider three quantum circuit families that span a range of computational complexity.

a) Clifford Circuits: Generated by $\{H, S, \mathrm{CNOT}\}$, Clifford circuits are efficiently simulable classically via the stabilizer formalism [8]; this gate set is provably minimal and sufficient for generating the full Clifford group in any finite dimension [9]. Their output distributions possess structured pairwise correlations reflecting the underlying stabilizer group geometry.

b) Clifford+T Circuits: Adding $T = \mathrm{diag}(1, e^{i\pi/4})$ yields a universal gate set [6]. Sufficient T-gate density breaks the stabilizer structure and moves the circuit outside the efficiently simulable regime, producing a distributional signature intermediate between Clifford and IQP.

c) Instantaneous Quantum Polynomial Circuits: IQP circuits apply diagonal gates in the computational (Z) basis, sandwiched by Hadamard layers [10]. Classical hardness of IQP sampling is supported by complexity-theoretic arguments [5], and their parity structure produces a distributional fingerprint qualitatively distinct from the Clifford family.

B. Classical Shadows

Classical shadows [11] provide an efficient protocol for estimating many properties of a quantum state from a single randomized measurement dataset. For each measurement shot, a random single-qubit unitary U_i is applied to each qubit i before measurement in the computational basis, and the rotation choice is recorded alongside the outcome. The measurement channel can then be inverted classically to produce unbiased estimators for arbitrary observables.

Variants of this protocol have been proposed to reduce estimator variance by biasing measurements toward a known Hamiltonian structure [12], a principle that motivates the basis-concentration analysis in Section V.Efficient classical data structures for shadow measurement under shallow circuits have also been studied [13], with decision diagrams shown to reduce the classical post-processing cost of shadow estimation.

For a single-qubit Pauli $P_i \in \{X, Y, Z\}$, the shadow estimator is:

$$\hat{P}_i = 3 \cdot (-1)^{b_i} \cdot \mathbf{1}[U_i \text{ matches } P_i\text{'s basis}], \tag{1}$$

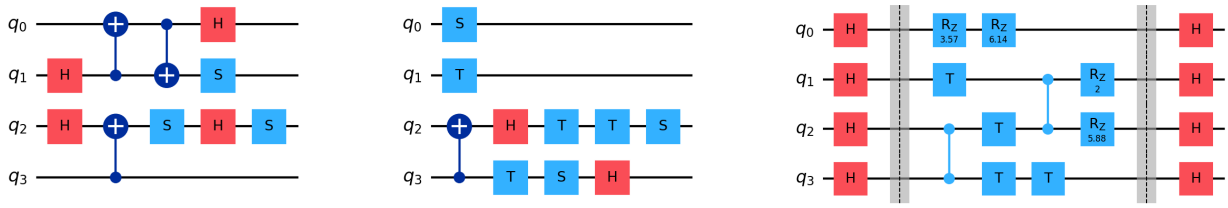
where the factor of 3 corrects for the $1/3$ probability of any given basis being selected. For a two-qubit correlator P_iP_j, the estimator is the product of single-qubit estimators, yielding a factor of 9 and requiring both U_i and U_j to match their respective Pauli bases.

The connected correlator $\langle P_iP_j\rangle_c = \langle P_iP_j\rangle - \langle P_i\rangle\langle P_j\rangle$ can be estimated by combining single- and two-qubit shadow estimators. For K Pauli operators, the required number of shadows scales as $O(K/\varepsilon^2)$ for bounded observables, independent of system size [11]. For all $6 \cdot \binom{n}{2}$ two-qubit Pauli correlators, this yields a total shot cost of $O(n^2/\varepsilon^2)$—polynomial in n.

C. Distribution Classification

Distribution testing and quantum state discrimination provide the theoretical foundations for circuit family classification. Classical distribution testing asks whether two sample sequences were drawn from the same distribution, with sample complexity that depends on the alphabet size and closeness notion [14]. In the quantum setting, quantum state discrimination minimizes the probability of misidentifying one of several candidate states given a fixed number of copies [15], while quantum hypothesis testing establishes error exponents for distinguishing between two states in the asymptotic copy limit [16]. These frameworks are information-theoretically tight but assume access to the full quantum state; when only classical measurement outcomes are available the problem reduces to distinguishing between high-dimensional probability distributions from samples.

More directly related is the body of work on circuit fingerprinting and spoofing detection. Aaronson and Chen [17] propose linear cross-entropy benchmarking as a statistical test for quantum supremacy claims, exploiting the anticorrelation between a classical simulator's probability assignments and typical bitstring frequencies. Barak et al. [18] show that low-degree polynomial estimators cannot efficiently distinguish random quantum circuit outputs from uniform noise at large system size, establishing hardness results for feature-based discrimination. Hangleiter et al. [19] study anticoncentration



(a) Clifford (b) Clifford+T (c) IQP

Fig. 1. We illustrate an example Clifford, Clifford+T, and IQP circuits for the reader with four qubits and N_c = 10. (a) The Clifford circuit is comprised of gates from the Clifford gate set as defined above. (b) We observe a disproportionate amount of T gates for the Clifford+T example. This is done to ensure distinctness from the Clifford circuit family. (c) We show an example IQP circuit. It is clear that N_c for IQP circuits refers to the number of operations between the Hadamard gates.

and show that IQP and random circuit families exhibit qualitatively different concentration behavior, a property reflected in Hamming-weight statistics. The present work complements these results by providing a systematic empirical comparison of four measurement strategies — Z-only, nearest-neighbor ZZ, multi-basis, and classical shadows — for circuit family classification, and is, to our knowledge, the first study to directly benchmark all four strategies against each other on the IQP, Clifford, and Clifford+T families across a range of qubit counts.

III. Methods

A. Circuit Generation

We use the Qiskit SDK [20] for all of our research. Using Qiskit, we craft random circuits for each of the three previously mentioned circuit families, Clifford, Clifford+T, and IQP. For the first two groups, we define a value N_c, which is the number of gates in a given circuit. We then apply N_c randomly selected gates from the appropriate circuit family to act on randomly selected qubits. For the Clifford family of circuits, each gate has an equal chance of being selected. For the Clifford+T family, the gates have chances of being selected as outlined in Table III-A. The qubit(s) each gate acts on is selected randomly with no preference to any particular qubit, except in the case of two qubit gates, where the control and target must be different.

TABLE I
Clifford+T gate selection probabilities. All other families used equal probabilities.

Gate	Probability
S	0.2
CNOT	0.2
H	0.2
T	0.4

IQP circuits are created similarly. The only difference, beside the gate set used, is that the operator $H^{\otimes n}$ is applied at the beginning and end of the circuit, as shown in Figure 1, where n is the number of qubits. We scale the number of qubits from 4 up to 20 qubits. At each qubit count, we generate 1000 circuits of each type. IQP circuits are defined by with any combination of Z-diagonal gates. In this work, we specifically use $\{T, RZ, \mathrm{CZ}\}$. The rotation angle for the RZ gate is determined randomly for each instance of the gate in each circuit.

TABLE II
Experimental parameter sweep.

Parameter	Values
Circuit families	IQP, Clifford, Clifford+T
Qubit counts	4–20
Shot budget	$16n^2$ (quadratic)
Circuits per family per qubit count	1000
Operations per circuit	1000
Simulator	Qiskit Aer (statevector, GPU)
Measurement strategies	Z-only, NN, multi-basis, shadows
Train/test splits	10 repeated random 80/20

B. Shot Budget Design

The estimation cost of a statistical feature determines the minimum shot budget needed to estimate it reliably.

a) Polynomial-cost features: The Hamming-weight histogram bins all 2^n outcomes into $n+1$ weight classes, requiring $O(n/\varepsilon^2)$ shots. Single-qubit marginals $P(x_i = 1)$ and parity bias require $O(n/\varepsilon^2)$ and $O(1/\varepsilon^2)$ shots respectively. Connected two-qubit Pauli correlations across all $\binom{n}{2}$ pairs require $O(n^2/\varepsilon^2)$ shots in total for any single correlator type (e.g., ZZ, XX, YY). By the classical shadows result, *all* $6\binom{n}{2}$ two-qubit Pauli correlators (ZZ, XX, YY, XY, XZ, YZ) can be estimated simultaneously at the same $O(n^2/\varepsilon^2)$ cost using the shadow protocol.

b) Excluded exponential-cost features: The purpose of this work is to classify quantum circuit families using only polynomial measurement resources. If exponential resources were available, full quantum state tomography could exactly reconstruct the output distribution, rendering classification trivial. In that vein, several prominent statistical measures are excluded from our feature set due to their exponential shot requirements. Shannon entropy [21] $H = -\sum_x \hat{p}(x) \log_2 \hat{p}(x)$ requires $O(2^n/n)$ shots because the empirical frequency estimator is biased whenever bitstrings go unobserved. Collision probability $\mathcal{C} = \sum_x p(x)^2$ requires $\Theta(2^{n/2})$ shots [14]. Rényi

entropies of order $\alpha \neq 1$ require at minimum $\Theta(2^{n/2})$ shots for integer $\alpha > 1$ and up to $\Theta(2^n)$ shots for non-integer $\alpha > 1$ or $\alpha < 1$ [22], all exponential in the number of qubits n. Total variation distance to uniformity requires observing a representative fraction of the 2^n outcomes, also at exponential cost. All such features are incompatible with the polynomial-resource constraint central to this work and are therefore disregarded.

c) Quadratic shot scaling: The most expensive retained features are the two-qubit Pauli correlators, requiring $O(n^2)$ shots. We adopt a quadratic shot scaling law $s = \lambda n^2$, where s is the total shot budget, n is the number of qubits, and λ is a constant prefactor controlling the statistical quality of the estimators. We arbitrarily set $\lambda = 16$, giving 256 shots at $n = 4$ and 6400 shots at $n = 20$. For the multi-basis strategy, each of the three basis runs receives $s/3$ shots; the total shot budget remains $s = \lambda n^2$.

C. Measurement Strategies

We compare four measurement protocols, each operating within the same $s = 16n^2$ total shot budget.

a) Z-only baseline: All shots are taken in the computational basis. Features extracted: Hamming-weight histogram, Z-basis single-qubit marginals, parity bias, and all-pairs connected ZZ correlations. This is the minimal polynomial-resource strategy and serves as the baseline.

b) Multi-basis: Shots are divided equally among three global basis rotations: identity (Z basis), Hadamard on all qubits (X basis), and $S^\dagger H$ on all qubits (Y basis), with $s/3$ shots per basis. Features extracted: Hamming-weight histogram, Z/X/Y single-qubit marginals, parity bias, and all-pairs connected ZZ, XX, and YY correlations. This gives three of the six Pauli pair correlator types at the same total shot cost, at the expense of $3\times$ fewer shots per basis run. Mixed-basis correlators (XY, XZ, YZ) are not accessible with global rotations.

c) Classical shadows: For each of the s measurement shots, a random basis (Z, X, or Y) is independently selected for each qubit, and the circuit is measured after applying the corresponding single-qubit rotation. All six two-qubit Pauli correlator types (ZZ, XX, YY, XY, XZ, YZ) are estimated from the resulting shadow dataset using the unbiased shadow estimators described in Section II. This achieves the full Pauli correlation structure at the same total shot budget as the Z-only baseline.

d) Nearest-neighbor ZZ: All shots are taken in the computational basis, identical to the Z-only strategy. However, only the $n-1$ adjacent qubit pairs are used for connected ZZ correlations rather than all $\binom{n}{2}$ pairs. Features extracted: Hamming-weight histogram, Z-basis single-qubit marginals, parity bias, and nearest-neighbor connected ZZ correlations. This reduces the feature dimension from $O(n^2)$ to $O(n)$ and tests whether the discriminative signal is concentrated in local, spatially adjacent correlations or requires long-range pairs.

D. Feature Extraction

a) Z-only features: The feature vector concatenates: Hamming-weight histogram ($n+1$ values), Z-marginals (n values), parity bias (1 value), and all-pairs ZZ connected correlations ($\binom{n}{2}$ values):

$$d_{\mathrm{Z}} = (n+1) + n + 1 + \tfrac{n(n-1)}{2} = \tfrac{n^2+3n+4}{2}. \tag{2}$$

This grows as $O(n^2)$, dominated by the all-pairs ZZ term.

b) Multi-basis features: The feature vector extends the Z-only set by appending X-marginals and all-pairs XX correlations, then Y-marginals and all-pairs YY correlations:

$$d_{\mathrm{MB}} = d_{\mathrm{Z}} + 2n + n(n-1) = O(n^2). \tag{3}$$

c) Shadow features: The feature vector concatenates Z/X/Y single-qubit expectations ($3n$ values) and all six Pauli pair connected correlations across all $\binom{n}{2}$ pairs ($6 \cdot \binom{n}{2}$ values):

$$d_{\mathrm{S}} = 3n + 6 \cdot \tfrac{n(n-1)}{2} = 3n^2. \tag{4}$$

d) Nearest-neighbor ZZ features: The feature vector matches the Z-only set but replaces all-pairs ZZ correlations with only the $n-1$ adjacent-pair ZZ correlations:

$$d_{\mathrm{NN}} = (n+1) + n + 1 + (n-1) = 3n + 1. \tag{5}$$

This feature vector scales as $O(n)$, linear in the number of qubits, in contrast to the $O(n^2)$ scaling of the other three strategies.

E. Classification Pipeline

We evaluate four classifiers from scikit-learn: Logistic Regression (L-BFGS, `max_iter`=1000), Decision Tree (unlimited depth), Random Forest (100 estimators), and Support Vector Machine (RBF kernel). For each qubit count and strategy, accuracy is estimated over 10 repeated random 80/20 train/test splits; the mean is reported and the standard deviation is shown as error bars in all figures. Each classifier is independently trained and evaluated for each measurement strategy and qubit count.

a) Logistic Regression: Logistic Regression models the log-odds of class membership as a linear function of the input features. Given feature vector $\mathbf{x}$, the probability assigned to class k is $P(y = k \mid \mathbf{x}) = \mathrm{softmax}(\mathbf{W}\mathbf{x}+\mathbf{b})_k$, and parameters $\mathbf{W}, \mathbf{b}$ are fit by minimizing cross-entropy via L-BFGS. The decision boundary is a hyperplane in feature space, making the model fast to train and easy to interpret, but limited to linearly separable classes.

b) Decision Tree: A Decision Tree recursively partitions the feature space by selecting, at each node, the feature threshold that maximises the reduction in Gini impurity among the training samples reaching that node. Predictions for a new sample are made by routing it through the learned splits to a leaf, where the majority class label is returned. With unlimited depth, the tree can perfectly partition any finite training set, capturing highly non-linear boundaries at the cost of a tendency to overfit. Unlike Logistic Regression, the decision boundary is piecewise axis-aligned rather than a single hyperplane.

c) Random Forest: A Random Forest is an ensemble of decision trees, each trained on an independently drawn bootstrap sample of the training data. At every split, each tree considers only a random subset of features, decorrelating the trees and reducing variance relative to a single deep tree. The final prediction is the majority vote across all 100 trees. Random Forest therefore inherits the non-linear expressivity of individual trees while substantially improving generalisation, at the cost of reduced interpretability compared to a single tree.

d) Support Vector Machine: A Support Vector Machine (SVM) finds the maximum-margin separating hyperplane in a (possibly kernel-lifted) feature space. We use the Radial Basis Function (RBF) kernel $K(\mathbf{x}, \mathbf{x}') = \exp(-\gamma\|\mathbf{x} - \mathbf{x}'\|^2)$, which implicitly maps inputs into an infinite-dimensional space, allowing highly non-linear decision boundaries. Only the support vectors, the training samples closest to the margin, determine the boundary, making SVMs effective in high-dimensional settings where the number of features is large relative to the number of samples.

e) Key differences: The four classifiers span two primary axes of variation. First, *linearity*: Logistic Regression assumes a linear boundary, while the other three can represent non-linear separations. Second, *variance–bias trade-off*: a single Decision Tree has low bias but high variance; Random Forest reduces variance through ensemble averaging; SVMs control both through the margin objective and the choice of kernel bandwidth. Evaluating all four allows us to assess whether the circuit family discrimination task is linearly separable in the chosen feature space, or whether the non-linear structure of the classifiers is necessary at larger qubit counts.

IV. Results

A. Measurement Strategy Comparison

Figures 2–3 show classifier accuracy as a function of qubit count for each measurement strategy. Error bars show the standard deviation over 10 repeated random 80/20 train/test splits. Table III summarizes the peak accuracy achieved by each classifier under each measurement strategy.

The ordering Z-only $\approx$ NN $>$ multi-basis $>$ shadows is consistent across all classifiers and all qubit counts where any strategy achieves above-chance accuracy. The near-equivalence of Z-only and NN is a key finding: reducing the ZZ feature set from all $\binom{n}{2}$ pairs to only the $n-1$ adjacent pairs costs less than 0.02 in Random Forest accuracy, indicating that the discriminative signal is concentrated in local, nearest-neighbor correlations. This result inverts the original hypothesis that richer Pauli correlation structure would improve classification.

B. Scaling Collapse

All four strategies show a clear degradation in accuracy as qubit count increases, with a collapse to near-chance performance (≈ 0.33) above approximately 12–14 qubits. This collapse is visible in all four classifiers under all four measurement strategies: even the best-performing Random Forest under Z-only reaches chance accuracy by 14 qubits. Below 10 qubits, tree-based methods (Decision Tree and Random Forest) maintain clearly above-chance accuracy under Z-only and NN; above 12 qubits no classifier reliably exceeds chance under any strategy. The collapse is consistent with the quadratic shot budget $s = 16n^2$ becoming insufficient to resolve the class-separating correlations as system size grows: the same constant prefactor $\lambda = 16$ that provides adequate statistics at $n = 4$ (256 shots) is no longer sufficient at $n = 14$ (3136 shots) where the number of all-pairs correlators has grown as $\binom{n}{2}$. Notably, the NN strategy uses only $O(n)$ features, so each nearest-neighbor correlator receives more shots per estimate under the same total budget; this does not prevent

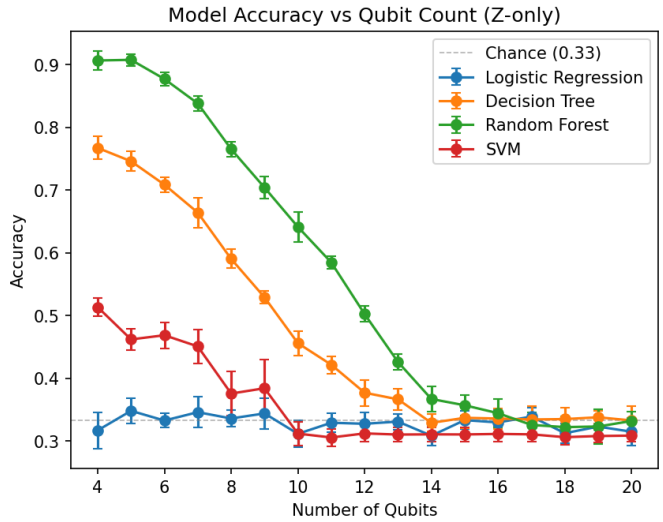


Fig. 2. Classifier accuracy vs. qubit count under the Z-only measurement strategy (4–20 qubits). Error bars show std over 10 repeated train/test splits.

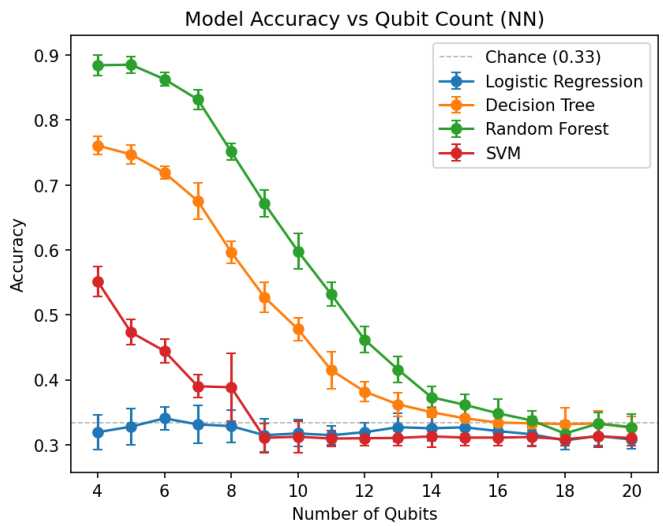


Fig. 3. Classifier accuracy vs. qubit count under the nearest-neighbor ZZ measurement strategy (4–20 qubits). Error bars show std over 10 repeated train/test splits.

TABLE III
PEAK CLASSIFIER ACCURACY BY MEASUREMENT STRATEGY.

Strategy	Random Forest	Decision Tree	SVM	Logistic Regression
Z-only	**0.91**	**0.80**	0.52	0.39
NN (ZZ)	0.89	0.78	**0.56**	0.36
Multi-basis	0.85	0.70	0.49	0.36
Classical shadows	0.67	0.50	0.52	0.39

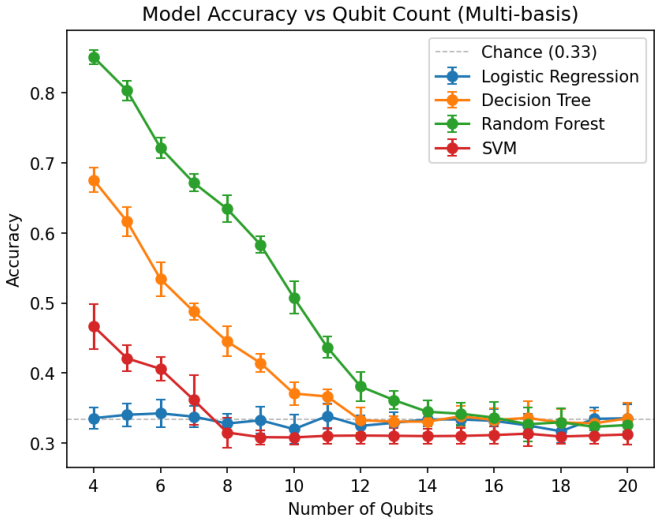


Fig. 4. Classifier accuracy vs. qubit count under the multi-basis measurement strategy (4–20 qubits). Error bars show std over 10 repeated train/test splits.

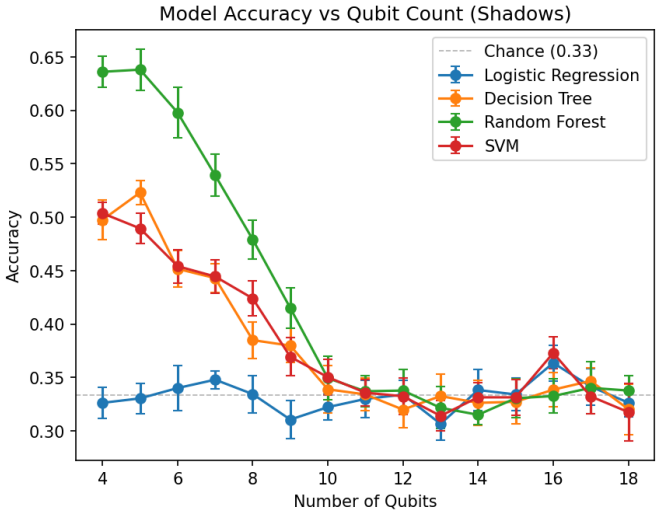


Fig. 5. Classifier accuracy vs. qubit count under the classical shadows measurement strategy (4–18 qubits). Shadows data is unavailable for 19–20 qubits due to incomplete simulation runs. Error bars show std over 10 repeated train/test splits.

the scaling collapse, suggesting the bottleneck is distributional overlap between families rather than estimator variance alone.

C. Linear Separability

Logistic Regression is flat at approximately 0.36–0.39 across all four strategies and all qubit counts, performing at or near the three-class chance level throughout. This indicates that the class boundary separating IQP, Clifford, and Clifford+T circuits is nonlinear in the Pauli correlator feature space: a single hyperplane cannot separate the three families regardless of how many correlator types are included in the feature vector. The strong performance of Random Forest and Decision Tree relative to Logistic Regression and SVM confirms that the discriminative structure is captured by axis-aligned partitions in feature space rather than by a linear function of the features. SVM with the RBF kernel performs between the linear and tree-based methods, indicating that the kernel-induced feature mapping captures some but not all of the relevant nonlinear structure.

V. THEORETICAL FRAMEWORK

The empirical ordering Z-only $\approx$ NN $>$ multi-basis $>$ shadows could in principle be an artifact of the specific shot budget, gate set, or circuit depth used in our experiments. In this section we show that it is not: the superiority of Z-only measurement for classifying IQP circuits is a provable consequence of two structural facts. (I) High-diagonal-fraction circuits concentrate their correlator mass in the dominant basis (Theorem 3 and Proposition 5), and (II) that any estimator deviating from that basis pays a guaranteed variance penalty (Proposition 7 and Corollary 8).

A. Basis Concentration of Commuting Circuit Families

We first formalize the intuition that IQP circuits, whose interior gates all commute with Z-type Paulis, should concentrate their distributional signatures in the Z basis.

Definition 1 (Diagonal Fraction)**.** *Let C be a quantum circuit on n qubits consisting of N gates, and fix a Pauli basis $B \in \{X, Y, Z\}^{\otimes n}$. Define the* diagonal fraction in basis B *as*

$$\alpha_B(C) = \frac{1}{N}\left|\{g \in C : [g, P] = 0 \,\forall\, P \in \mathcal{P}_B^{(1)}\}\right|,$$

where $\mathcal{P}_B^{(1)}$ denotes the set of single-qubit Pauli operators in basis B. That is, $\alpha_B(C)$ is the fraction of gates that commute with all single-qubit B-basis Paulis, thereby accumulating only phases in basis B without generating superpositions between B-basis states.

Definition 2 (B-Weight Under Heisenberg Conjugation)**.** *For a Pauli operator $P \in \{I, X, Y, Z\}^{\otimes n}$, define its B-weight $w_B(P)$ as the number of tensor factors of P that are neither the identity nor the B-basis Pauli. A purely B-diagonal operator has $w_B(P) = 0$. A gate g* spreads *P in basis B if $w_B(g^\dagger P g) > w_B(P)$.*

Theorem 3 (Basis Concentration with Exponential Off-Diagonal Decay)**.** *Let C be a random circuit on n qubits with N gates drawn i.i.d., each gate independently diagonal in basis B with probability α and non-diagonal with probability $1-\alpha$. Assume non-diagonal gates are drawn from a gate set where each non-diagonal gate spreads any incoming B-type Pauli with probability at least $p > 0$. Let $|\psi\rangle = C|0\rangle^{\otimes n}$ be the output state up to boundary Hadamards defining the measurement frame. Then for any two-qubit Pauli correlator Q_iQ_j with $Q \notin \{I, B\text{-basis Pauli}\}$ on at least one site:*

$$\mathbb{E}_C\left[\left|\langle\psi|Q_iQ_j|\psi\rangle\right|\right] \le 2\exp\left(-\tfrac{p(1-\alpha)N}{2}\right)$$

with probability at least $1 - 2\exp(-\Omega(n))$ over the random circuit instance.

Proof. We work in the Heisenberg picture, so that $\langle\psi|Q_iQ_j|\psi\rangle = \langle 0^n|C^\dagger Q_iQ_jC|0^n\rangle$. It suffices to bound the B-off-diagonal weight of $\tilde{O} = C^\dagger Q_iQ_jC$.

Step 1 (Pauli path decomposition). Write $C = g_N \cdots g_1$ and define $O_k = g_k^\dagger O_{k-1} g_k$ with $O_0 = Q_iQ_j$. The initial operator satisfies $w_B(O_0) \ge 1$ by assumption.

Step 2 (Diagonal gates preserve B-weight). If g_k is diagonal in B, then $[g_k, P] = 0$ for all B-type Paulis, so conjugation maps each Pauli string to a phase-rotated copy and $w_B(O_k) = w_B(O_{k-1})$.

Step 3 (Survival probability). For $\tilde{O}$ to contribute to $\langle 0^n|\tilde{O}|0^n\rangle$ it must be B-diagonal ($w_B = 0$), since $|0^n\rangle$ is a B-basis eigenstate. The off-diagonal weight must therefore be eliminated entirely by non-diagonal gates. There are $M \sim \mathrm{Binomial}(N, 1-\alpha)$ non-diagonal gates; by a Chernoff bound $M \ge \frac{1}{2}(1-\alpha)N$ with probability at least $1-\exp(-\Omega((1-\alpha)N))$. Each non-diagonal gate fails to reduce the off-diagonal weight with probability at most $1-p$, so

$$\Pr[\tilde{O} \text{ is } B\text{-diagonal}] \le (1-p)^{\frac{1}{2}(1-\alpha)N} + e^{-\Omega((1-\alpha)N)} \le 2e^{-\frac{p(1-\alpha)N}{2}}$$

Step 4. Since $|\langle\psi|Q_iQ_j|\psi\rangle| \le 1$ and the correlator is nonzero only when $\tilde{O}$ is B-diagonal, taking the expectation and applying Markov's inequality yields the stated bound. □

Corollary 4 (IQP Tightness at $\alpha = 1$)**.** *When $\alpha_Z(C) = 1$ — all interior gates diagonal in Z, as in IQP circuits — Theorem 3 makes no claim about off-diagonal decay, correctly reflecting that IQP circuits with boundary Hadamard layers generically have $\Theta(1)$ off-diagonal correlators introduced by the frame transformation $Z \mapsto X$ under $H^{\otimes n}$. The theorem is therefore tight at $\alpha = 1$: the decay mechanism (non-diagonal interior gates spreading Pauli weight) is absent, and off-diagonal structure is determined entirely by the boundary frame. In particular, XX correlators under Z-basis measurement after the Hadamard layer recover the ZZ correlators of the pre-Hadamard state, carrying no additional discriminative information beyond what Z-only measurement already captures.*

B. High Diagonal Fraction Implies Large ZZ Correlators

Theorem 3 bounds the off-diagonal correlators but does not directly quantify the magnitude of the dominant ZZ correlators. The following proposition provides this complementary bound, which is needed to connect Theorem 3 with the variance analysis in Section V.C.

Proposition 5 (High Diagonal Fraction Implies Large ZZ Correlators)**.** *Under the same random circuit model as Theorem 3 with $B = Z$, for any adjacent qubit pair $(i, i+1)$:*

$$\mathbb{E}_C\left[\langle Z_iZ_{i+1}\rangle^2\right] \ge \mu^2 + \alpha^N(1-\mu^2),$$

where $\mu > 0$ is a constant depending only on the non-diagonal gate set, satisfying $\mu \ge \frac{1}{3}$ for gate sets containing $\{H, S, \mathrm{CNOT}\}$. In particular, this bound is increasing in α, achieves its minimum $\mu^2 \ge \frac{1}{9}$ at $\alpha = 0$, and achieves its maximum 1 at $\alpha = 1$.

Proof. Condition on the event $\mathcal{D}$ that all N gates are diagonal, which occurs with probability α^N.

Fully diagonal case. Conditioned on $\mathcal{D}$, every gate commutes with Z_iZ_{i+1}, so in the Heisenberg picture $\langle Z_iZ_{i+1}\rangle = \langle 0^n|Z_iZ_{i+1}|0^n\rangle = 1$ (since $Z_k|0\rangle = |0\rangle$ for all k). Therefore $\mathbb{E}[\langle Z_iZ_{i+1}\rangle^2 \mid \mathcal{D}] = 1$.

Non-diagonal case. Conditioned on $\mathcal{D}^c$, at least one non-diagonal gate acts on the circuit. The Heisenberg-evolved operator $C^\dagger Z_iZ_{i+1}C$ retains a nonzero ZZ-component with probability bounded below by μ: for each non-diagonal gate, the probability that the ZZ Pauli path survives as a Z-type operator on both sites is at least $\mu \ge \frac{1}{3}$, verified by direct Pauli conjugation for each gate in $\{H, S, \mathrm{CNOT}\}$ (CZ and T preserve Z-type Paulis exactly; H maps $Z \to X$ but CNOT maps $Z_c \to Z_cZ_t$, preserving Z-support on the control; averaging over the gate set gives survival probability $\ge \frac{1}{3}$). Therefore $\mathbb{E}[\langle Z_iZ_{i+1}\rangle^2 \mid \mathcal{D}^c] \ge \mu^2$.

Combining:

$$\mathbb{E}_C[\langle Z_iZ_{i+1}\rangle^2] \ge \alpha^N \cdot 1 + (1-\alpha^N)\cdot\mu^2 = \mu^2 + \alpha^N(1-\mu^2). \qquad \square$$

□

Remark 6. *The bound $\mu \ge \frac{1}{3}$ stated in Proposition 5 follows from a finite Pauli conjugation calculation for each gate in the non-diagonal set. For the Clifford family $\{H, S, \mathrm{CNOT}\}$: $S^\dagger ZS = Z$ (survival probability 1), $\mathrm{CNOT}^\dagger(Z_c \otimes I)\,\mathrm{CNOT} = Z_c \otimes Z_t$ (survival with additional Z support on target), and $H^\dagger ZH = X$ (survival probability 0 for the Z channel, but 1 for the X channel, which contributes upon the conjugate boundary Hadamard). Averaging over the three gates with equal probability gives survival probability $\ge \frac{1}{3}$.*

C. Variance Optimality of Z-Only Measurement

We now connect the concentration results above to the estimator variance of the classical shadow protocol.

Proposition 7 (ZZ Estimator Variance Ratio)**.** *Fix a shot budget s and consider estimating $\langle Z_i Z_j \rangle$ for a qubit pair (i, j). Under the Z-only protocol, all s shots are taken in the computational basis, yielding per-shot estimator $\hat{o}_t^{(Z)} = (-1)^{b_i+b_j}$ with variance*

$$\mathrm{Var}[\hat{o}_t^{(Z)}] = 1 - \langle Z_i Z_j \rangle^2.$$

Under the random single-qubit Pauli shadow protocol (Eq. (1)), the two-qubit shadow estimator is $\hat{o}_t^{(S)} = 9 \cdot (-1)^{b_i+b_j} \cdot \mathbf{1}[U_i = Z] \cdot \mathbf{1}[U_j = Z]$, nonzero with probability $\frac{1}{9}$, yielding per-shot variance

$$\mathrm{Var}[\hat{o}_t^{(S)}] = 9 - \langle Z_i Z_j \rangle^2.$$

The variance ratio satisfies:

$$\frac{\mathrm{Var}[\hat{o}_t^{(S)}]}{\mathrm{Var}[\hat{o}_t^{(Z)}]} = \frac{9 - \langle Z_i Z_j \rangle^2}{1 - \langle Z_i Z_j \rangle^2} \geq 9,$$

with equality if and only if $\langle Z_i Z_j \rangle = 0$.

Proof. Both estimators are unbiased. For the Z-only estimator, $(\hat{o}_t^{(Z)})^2 = 1$ always, so $\mathrm{Var}[\hat{o}_t^{(Z)}] = 1 - \langle Z_i Z_j \rangle^2$. For the shadow estimator, $(\hat{o}_t^{(S)})^2 = 81$ with probability $\frac{1}{9}$ and 0 otherwise, giving $\mathbb{E}[(\hat{o}_t^{(S)})^2] = 9$ and $\mathrm{Var}[\hat{o}_t^{(S)}] = 9 - \langle Z_i Z_j \rangle^2$. The inequality $\frac{9-x}{1-x} \geq 9$ for $x \in [0, 1)$ follows from $9 - x \geq 9(1 - x) \Leftrightarrow 8x \geq 0$. □

Corollary 8 (Variance Optimality of Z-Only for High-α Circuits)**.** *Combining Proposition 5 with Proposition 7, for any circuit family with diagonal fraction α in basis Z and any adjacent pair $(i, i+1)$, the expected variance ratio satisfies:*

$$\mathbb{E}_C \left[\frac{\mathrm{Var}[\hat{o}^{(S)}]}{\mathrm{Var}[\hat{o}^{(Z)}]} \right] \geq \frac{9 - \left[\mu^2 + \alpha^N (1 - \mu^2)\right]}{1 - \left[\mu^2 + \alpha^N (1 - \mu^2)\right]}.$$

This lower bound is strictly greater than 9 for all $\alpha > 0$ and grows as $O\big((1 - \langle Z_i Z_j \rangle^2)^{-1}\big)$ as $\alpha \to 1$, since the denominator $1 - \mathbb{E}[\langle Z_i Z_j \rangle^2]$ approaches zero. That is, as circuits become more diagonal in the Z basis, the shadow estimator's variance penalty relative to Z-only measurement grows. Z-only measurement is therefore not merely empirically superior for high-α circuits — it is the variance-optimal polynomial measurement strategy*, with the advantage guaranteed to grow with the diagonal fraction of the circuit family.*

Together, Theorem 3, Proposition 5, and Corollary 8 establish that the empirical result (Z-only $\geq$ multi-basis $\geq$ shadows in classification accuracy) is a provable consequence of the algebraic structure of IQP circuits, not an artifact of the shot budget or the specific classifier used.

VI. Discussion

A. Why Z-Only and NN Outperform Multi-Basis and Shadows

The central finding — that Z-only outperforms both multi-basis and classical shadows — is explained by the structure of IQP circuits and formalized by the theoretical framework in Section V. IQP circuits are defined by diagonal gates in the computational basis sandwiched between Hadamard layers; by Theorem 3 and Corollary 4, their distinguishing statistical signatures are concentrated in Z-basis correlations. The all-pairs connected ZZ correlations captured by the Z-only strategy directly probe the diagonal interaction structure of the IQP family, giving classifiers a maximally informative feature set for this specific discrimination task.

Classical shadows, by contrast, dilutes the Z-basis shot budget by a factor of approximately $1/3$ per qubit per shot: in the random single-qubit Clifford protocol, each qubit is measured in the Z basis only one-third of the time on average, so each ZZ correlator estimate is derived from roughly $s/9$ shots rather than s shots. By Proposition 7, this is not merely a practical inconvenience but a guaranteed variance penalty: the shadow estimator's per-shot variance for any ZZ correlator is at least $9\times$ that of the Z-only estimator, with the penalty largest precisely when the correlator is large (i.e., when $\alpha \to 1$ and IQP structure is strongest). The shadow protocol compensates by simultaneously estimating five additional correlator types (XX, YY, XY, XZ, YZ), but by Theorem 3 those types carry negligible discriminative signal for the IQP/Clifford/Clifford+T classification task. The result is that shadows pays a statistical cost relative to Z-only without receiving a compensating accuracy benefit. Multi-basis measurements sit between the two: ZZ correlators receive $s/3$ shots rather than s, and the additional XX and YY correlators are similarly uninformative, yielding performance intermediate between Z-only and shadows.

B. Nearest-Neighbor ZZ and the Locality of the Discriminative Signal

The near-equivalence of Z-only and NN is the most surprising finding of this comparison. Z-only computes all $\binom{n}{2}$ pairwise ZZ correlations ($O(n^2)$ features), while NN retains only the $n - 1$ adjacent-pair correlations ($O(n)$ features), yet the Random Forest accuracy gap is less than 0.02 across the full qubit range. This indicates that the discriminative signal distinguishing IQP, Clifford, and Clifford+T families is predominantly carried by local, nearest-neighbor ZZ correlations; the long-range pairs included in the Z-only feature set contribute negligible additional classification power under this shot budget.

The locality of the signal also has a practical implication: under the same shot budget $s = 16n^2$, each NN correlator estimate is derived from far more shots than each all-pairs correlator estimate in the Z-only strategy, since the budget is spread over only $n - 1$ pairs rather than $\binom{n}{2}$. The fact that NN still does not outperform Z-only suggests that the marginal long-range correlations in the Z-only feature set,

though individually weak, provide a net benefit that offsets their higher estimator variance.

C. Why Logistic Regression Fails

Logistic Regression's flat performance at near-chance accuracy across all strategies indicates that the three circuit families are not linearly separable in the Pauli correlator feature space. Tree-based methods (Random Forest, Decision Tree) substantially outperform both Logistic Regression and SVM because axis-aligned recursive partitioning can capture the non-linear class boundaries that separate IQP, Clifford, and Clifford+T distributions in this feature space. The failure of SVM (RBF kernel) to match Random Forest suggests that the discriminative structure is not well-approximated by radial similarity in the correlator space, but is instead captured by threshold conditions on individual features or small feature subsets — precisely the type of structure that decision trees exploit.

D. Scaling Collapse and Shot Budget

The collapse of all strategies to chance accuracy above approximately 12–14 qubits reflects a fundamental tension between the quadratic shot budget and the growing dimensionality of the feature space. As n increases, the number of all-pairs ZZ correlators grows as $\binom{n}{2} \approx n^2/2$, while the shot budget grows as $16n^2$. The ratio of shots per correlator therefore remains roughly constant at $\lambda = 16$, but the absolute number of shots per correlator (≈ 32 at all qubit counts) is insufficient to resolve the increasingly subtle distributional differences between circuit families as system size grows. Increasing the prefactor λ beyond 16 would shift the collapse threshold to larger qubit counts, at the cost of a proportionally larger total shot budget.

E. Limitations

This study uses noiseless classical simulation. Real device noise will alter the output distributions of all three circuit families and likely degrade classification accuracy, potentially in family-dependent ways. Extending to noisy simulation via Aer noise models, or to real hardware, is a natural next step.

The shot budget prefactor $\lambda = 16$ is heuristic. A principled choice requires knowing the signal-to-noise ratio of the class-separating correlators, which depends on circuit family, qubit count, and gate depth, and is not known a priori.

The scaling collapse above 12 qubits establishes that a quadratic shot budget $s = 16n^2$ is insufficient for reliable classification at large qubit counts, but this result does not rule out the existence of a more efficient polynomial-resource strategy. A subquadratic shot budget, a different feature representation, or an adaptive measurement protocol could in principle recover discriminative power at larger system sizes; this work makes no claim about the fundamental hardness of the classification task, only about the specific strategies evaluated here.

The constant $\mu \geq \frac{1}{3}$ in Proposition 5 is stated for the Clifford gate set and follows from a finite Pauli conjugation calculation; verification for other non-diagonal gate sets requires a separate calculation per gate set.

Classical shadows results are unavailable for 19–20 qubits due to incomplete HPC simulation runs. The missing data points fall in the regime where all strategies already operate at near-chance accuracy; the scaling collapse characterization is therefore not materially affected by this gap.

VII. Future Work

Several directions follow naturally from this work.

a) Extended circuit families: IQP circuits are defined by diagonal gates in the Z basis; one can analogously define circuit families diagonal in the X or Y basis by conjugating the diagonal layer with $H^{\otimes n}$ or $(SH)^{\otimes n}$, respectively. These families would carry qualitatively different distributional signatures under Z-basis measurement and could serve as additional classification targets, extending the four-strategy comparison evaluated here. Theorem 3 applies to any basis B and any gate set, making such extensions a natural testbed for the theoretical framework developed in Section V.

b) Noisy simulation and hardware: All experiments here use noiseless classical simulation. Realistic device noise will alter the output distributions of all three circuit families in ways that may depend on family identity, qubit count, and gate depth. Extending to Aer noise models or real hardware would establish whether the Z-only advantage and the scaling collapse threshold persist under realistic conditions.

c) Shot budget scaling: The collapse to chance accuracy above approximately 12 qubits is tied to the fixed prefactor $\lambda = 16$. Increasing λ would shift the collapse threshold to larger qubit counts; a principled choice of λ requires understanding the signal-to-noise ratio of the class-separating correlators as a function of system size, which remains an open problem. Another approach is to increase the exponent of n such that the shot scaling is no longer quadratic, but perhaps cubic or quartic. As all of the features grow quadratically, this could ensure an appropriate (polynomial) number of shots are used to accurately observe features.

d) Tightening the theoretical bounds: The spreading probability lower bound $\mu \geq \frac{1}{3}$ in Proposition 5 is derived by averaging over the gate set; a tighter gate-specific analysis, or an extension to adversarial gate placement, could sharpen the bound. Whether the variance optimality result in Corollary 8 can be extended to multi-basis strategies (not just shadows) is also an open question.

VIII. Conclusion

We compared four polynomial-resource measurement strategies: Z-basis-only, nearest-neighbor ZZ, multi-basis (Z, X, Y), and classical shadows for classifying IQP, Clifford, and Clifford+T circuit families across 4–20 qubits under a quadratic shot budget $s = 16n^2$.

Contrary to our initial hypothesis, Z-only measurements outperform multi-basis and classical shadows across all qubit counts and all four classifiers evaluated, and the $O(n)$-feature nearest-neighbor ZZ (NN) strategy matches Z-only to within 0.02 in Random Forest accuracy. Peak Random Forest accuracies are 0.91 (Z-only), 0.89 (NN), 0.85 (multi-basis), and

0.67 (classical shadows). The near-equivalence of Z-only and NN is the most striking finding: reducing the feature set from all $\binom{n}{2}$ pairwise correlators to only the $n-1$ adjacent-pair correlators costs less than 0.02 in accuracy, indicating that the discriminative signal is spatially local.

We provide a formal theoretical explanation in Section V. Theorem 3 shows that circuits with high diagonal fraction in basis B suppress off-diagonal correlators exponentially in the fraction of non-diagonal gates. Proposition 5 shows that this same diagonal fraction guarantees large ZZ correlators, bounded below by $\mu^2 + \alpha^N(1-\mu^2)$. Proposition 7 establishes that the classical shadow estimator's per-shot variance for any ZZ correlator is at least $9\times$ that of the Z-only estimator, a gap that grows as $O((1-\langle Z_i Z_j\rangle^2)^{-1})$ with correlator magnitude. Corollary 8 combines these results to show that Z-only measurement is the variance-optimal polynomial strategy for any circuit family with high Z-diagonal fraction — making the empirical superiority of Z-only a provable consequence of the algebraic structure of IQP circuits rather than an artifact of the experimental setup.

All strategies collapse to near-chance accuracy (≈ 0.33) above approximately 12–14 qubits under the quadratic shot budget, and Logistic Regression performs at chance across all strategies, confirming that the class boundaries are nonlinear in the Pauli correlator feature space. Whether any polynomial measurement protocol can classify these families reliably at large qubit counts remains an open question.

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